Python

A quickstart into the key concepts of programming

AI-powered (Python) coding

Why using AI for coding?

- The code legacy is used to train large language models
- The existing/legacy software examples are very rich and diverse
- AI can make reasonable recommendations, specially for generic programming, error interpretation, debugging etc
- AI lifts the weight of having-to-know-it-all from user's shoulders

- AI hallucinates! Your recommended code works but may produce garbage
- You are still the sole responsible for the outcome

Which backend to use?

- The choices of LLMs are ever-growing: Pick your favorite
- The well-known ones perform equally good for coding
- In this training, we use Microsoft Copilot (KU Leuven license)
- Also, Github belongs to Microsoft!
  Thus, Microsoft Copilot is well-trained

Where does AI help with programming?

- Suggesting good use of a language standard library or well-known 3rd parties. E.g.:
- Argument parsing
- Code styling
- Writing (inline) documentation and/or docstrings
- Writing unit tests (strongly relevant to capture possible mistakes)

Shop-n-Go vs. Interactive coding

Shop-n-Go

- Pitch your idea/project as a ‘prompt’
- AI generates a (Python) code for you
- You copy-paste and run the code

Interactive Coding

- You start a new file/project in IDE, and you give coding hints (e.g. AI chat)
- AI makes code-block suggestions or completions
- You proceed with finishing your project incrementally and interactively

Using Microsoft Copilot

Possible interfaces:

- Copilot on the web (using KU Leuven license): https://copilot.cloud.microsoft/
- Copilot app (login with KU Leuven/MFA)
- Copilot integration in Visual Studio Code
  See https://code.visualstudio.com/docs/copilot/overview

Setting up Microsoft Copilot in VSCode

Requirements and steps (see this link)

- Install Visual Studio Code: https://code.visualstudio.com/download
- (Optional) Create a Github account: https://github.com/
- (Recommended) Install Microsoft Python extension in VSCode
- In VSCode, at the bottom right, click on
- Authenticate using your KU Leuven account (MFA)
- Enable the chat: `Ctrl+Alt+I`

Demo 1: Correlated health metrics

- Question: Which health metrics correlate with one another?
- Dataset: From Kaggle https://www.kaggle.com/datasets/jockeroika/life-style-data/data Download as `.zip` file and extract in a working directory
- In VSCode: create `lifestyle-metrics.py`
- Make sure your Microsoft Copilot is ready
- Use comment lines `#` as prompts to guide the AI engine
- Accept suggestions with `<TAB>` or skip them with `<Enter>`
- Give prompts one by one: let AI suggest code incrementally

Demo 1: Correlated health metrics

Some inspirations:

```
# import necessary packages only from python standard
library
# Read CSV file and load data
# Convert data to column-major
# Extract column names and types; cast str -> float
# Select some numerical columns with health-metric
# Compute pairwise correlation between columns
# Print the correlation metrics
```

… and here be dragons!